

3. (a) An organisation has bought a software package containing several applications to be used in the various departments of the organisation.
- (i) State the name of this type of software package. (1 mark)
 - (ii) Give ONE advantage and ONE disadvantage of using this type of software package. (2 marks)
- (b) State the type of processing that is BEST suited for EACH of the following tasks:
- (i) Processing of monthly electricity bills (1 mark)
 - (ii) Processing of airline reservations (1 mark)
- (c) (i) Convert Octal 243 to decimal, binary and hexadecimal. (3 marks)
- (ii) Convert -856 to Binary Coded Decimal (BCD). (2 marks)
- Total 10 marks**

3) (a) (i) Integrated Software

(ii) Advantage : Data and information can be shared easily between these programs.

Disadvantage : It may not meet every ~~specific~~ ^{single} specific need of the user.

(b) (i) Batch Processing

(ii) Real-Time processing

$$\begin{aligned} \textcircled{c} (i) \quad 243_{10} &= 2 \times 8^2 + 4 \times 8^1 + 3 \times 8^0 \\ &= 2 \times 64 + 4 \times 8 + 3 \times 1 \\ &= 128 + 32 + 3 \\ &= 163_{10} \end{aligned}$$

$$\begin{aligned} \text{Decimal} \quad 243_{10} &= 2 \times 8^2 + 4 \times 8^1 + 3 \times 8^0 \\ &= 128 + 32 + 3 \\ &= 163_{10} \end{aligned}$$

Binary

$$\begin{array}{r|l} 2 & 163 \\ \hline 2 & 81 \text{ r } 1 \\ 2 & 40 \text{ r } 1 \\ 2 & 20 \text{ r } 0 \\ 2 & 10 \text{ r } 0 \\ 2 & 5 \text{ r } 0 \\ 2 & 2 \text{ r } 1 \\ 2 & 1 \text{ r } 0 \\ & 0 \text{ r } 0 \end{array}$$

$= 10100011_2$

Write the question number in this column.

Binary
Decimal

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2 | 163

2 | 81^{r1}

2 | 40^{r1}

2 | 20^{r0}

2 | 10^{r0}

2 | 5^{r0}

2 | 2^{r1}

2 | 1^{r0}

0^{r1}

210100011₂

A Hexadecimal

10100011

A 3

A3

0001

1

0010

2

11

3

100

4

101

5

110

6

111

7

1000

8

1001

9

1010

A

B

C

D

E

FP

(ii) 856

1011

1000

1001

0110

1011

1000

1001

0110

CSEC Information Technology

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Comments

Question 3 (a)

The question tested the candidate's ability to distinguish between systems and application programs and the advantages and disadvantages of the use of various types of application programs.

For Part (i), the candidate stated the correct type of software application used in the scenario given.

In Part (ii), the candidate demonstrated a good understanding of the advantages and disadvantages of the software package given in the scenario.

Question 3 (b)

This question tested the candidate's knowledge of different types of processing modes.

For both Parts (i) and (ii), the candidate stated the correct processing modes used in the scenarios given.

Question 3 (c)

This question tested the candidate's ability to manipulate the binary, decimal, octal and hexadecimal number systems.

In Part (i), the candidate demonstrated the ability to correctly convert from octal to decimal, hexadecimal and binary.

In Part (ii), the candidate demonstrated the ability to correctly convert to BCD.

Summary

The candidate's performance on the question was excellent.